

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I N. Jaffrin / 192521200 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Jaffrin

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty
Assessment Tasks

Total Marks Awarded:

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Assessment Task - 1

1) Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.

* cloud computing evolved through several stages of technological advancement. Initially, standalone computing systems were used, where applications and data were stored and processed on a single computer. These systems had limited storage, processing power.

* The next stage was client-server computing, where multiple client computers connected to a centralized server. This improved resource sharing, data management, and collaboration.

* With the growth of the Internet, distributed computing emerged. Multiple computers worked together over a network to perform tasks and share resources. Technologies such as grid computing and cluster computing further enhanced processing capabilities by combining the power of multiple machines.

* A major breakthrough was virtualization technology, which allowed multiple virtual machines to run on a single physical server.

virtualization improved resource utilization.

* The development of high-speed internet, web services, broadband networks, and data centers paved the way of cloud computing. cloud providers began offering computing resources.

* Today, cloud computing provide scalable, on-demand, and cost-effective services through models such as infrastructure as a service (IaaS), platform as a service (PaaS) and software as a service (SaaS).

Eg: Amazon web services, google cloud Platform (GCP)

Key Technological Transitions

Stage	Key Technology	Contribution
Standalone computing	Personal Computers	Local processing and storage.
Client-server computing	Network servers	Resource sharing and centralized.
Distributed management	Internet & Networking	Collaboration among multiple systems.
Grid / cluster computing	Parallel processing	Improved performance and scalability.

Conclusion:

The evolution from standalone computing to cloud-based systems was driven by advances in networking, distributed computing, virtualization. These transitions enabled organizations to access computing resources anytime and anywhere.

2) Interpret the working of Hypervisors in Virtualization. Differentiate between Type 1 and Type 2 Hypervisors with Example.

⇒ A hypervisor is software that enables virtualization by creating and managing multiple virtual machines (VMs) on a single physical computer. It acts as an intermediary between the hardware and the virtual machines.

Working of a Hypervisor.

- 1) The hypervisor is installed on a physical system.
- 2) It divides the hardware resource among multiple virtual machines.
- 3) Each VM runs its own operating system independent.
- 4) The hypervisor monitors and manages resource allocation to ensure efficient performance and

isolation between VMs.

Types of hypervisors.

Feature	Type 1 hypervisor (Bare-metal)	Type 2 hypervisor (Hosted)
Installation	directly on hardware	on top of a host operating system.
Performance	High	Moderate.
Security	more secure	less secure.
Resource usage	Efficient	Additional overhead due to host OS.
Usage	Enterprise data centers	Personal computers and testing.

Examples:

* Type 1 hypervisors: VMware ESXi, Microsoft Hyper-V.

* Type 2 hypervisors: Oracle VirtualBox, VMware Workstation, Parallels Desktop.

Conclusion:

Type 1 hypervisors provide better performance and scalability for cloud environments, while Type 2 hypervisors are suitable for development, testing, and educational purposes.

Describe the Role of APIs in cloud service Integration. Explain how APIs support Interoperability between services.

⇒ An application Programming Interface (API) is a set of rules and protocols that allows different software applications to communicate and exchange data. In cloud computing, APIs play a critical role in connecting cloud services.

Role of APIs in cloud service Integration.

- * Enable Communication between cloud application and services.

- * Allow developers to access cloud resources programmatically.

- * Support automation of cloud operations.

- * Facilitate data sharing across multiple platforms.

How APIs support Interoperability.

Interoperability refers to the ability of different systems and services to work together seamlessly.

APIs support Interoperability by:

- 1) Providing standardized communication methods.

- 2) Enabling data exchange between different cloud providers.

- 3) Allowing applications developed in different programming languages to interact.

4) Supporting integration of SaaS, PaaS, IaaS services

5) Enabling microservices architecture where independent services communicate efficiently.

Examples:

- * A mobile app using Google Maps API for location services.

- * An e-commerce website integrating PayPal API for payments.

- * Cloud applications using AWS APIs to manage storage and compute resources.

Conclusion:

APIs are the backbone of cloud integration, enabling seamless communication, automation, and interoperability between diverse cloud services and applications.

4) Explain the concept of Resource Pooling in cloud computing. Illustrate how it benefits multiple users.

⇒ Resource pooling is a cloud computing characteristic where computing resources such as servers, storage, memory, and network bandwidth are shared among multiple users through a common pool.

Working of Resource Pooling :

- 1) Resources are collected into a centralized pool.
- 2) Multiple users access resources from the same pool.
- 3) Virtualization separates user workloads.
- 4) Resources are dynamically assigned and reassigned based on demand.

Benefits to Multiple Users :

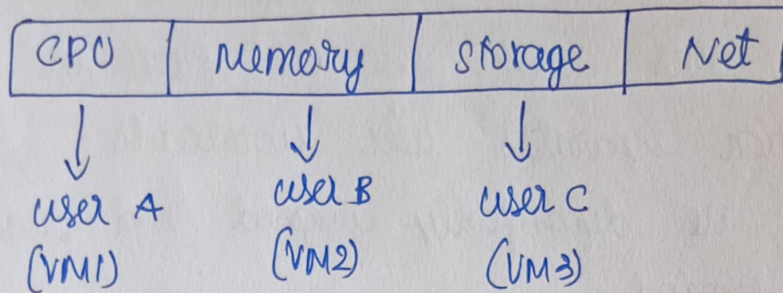
Benefit	Description.
Cost efficiency	User shares infrastructure costs.
Scalability	Resources can be increased or decreased as needed.
Better Utilization	Maximizes usage of available resources.
Flexibility	Resources are allocated dynamically.
High Availability	Resources remain accessible even during failures.

Example.

Consider a cloud provider such as AWS. Thousands of customers use the same data center infrastructure. Though virtualization and resource pooling, each customer receives the required storage and computing power without owning physical servers.

Illustration:

cloud Resource Pool.



Conclusion:

Resource pooling enables efficient sharing of cloud resources among multiple users while maintaining isolation, scalability, and cost-effectiveness. It is key feature that makes cloud computing economical and highly scalable.

- 5) Interpret the concept of Rapid Elasticity in cloud computing. Explain how it supports dynamic workloads.
- Rapid Elasticity is one of the essential characteristics of cloud computing. It refers to the ability of a cloud system to quickly and automatically increase or decrease computing resources such as CPU, memory, storage.

Working of Rapid Elasticity.

- 1) The cloud continuously monitors resource usage.
- 2) When demand increases, additional resources are automatically allocated.

- 3) when demand decreases, excess resources are released.
- 4) users pay only for the resources they actually use.
- 5) This process occurs with minimal human intervention.

Examples:

* E-commerce websites: During festival sales such as Amazon Prime day or Flipkart Big Billion days, cloud resources automatically increase.

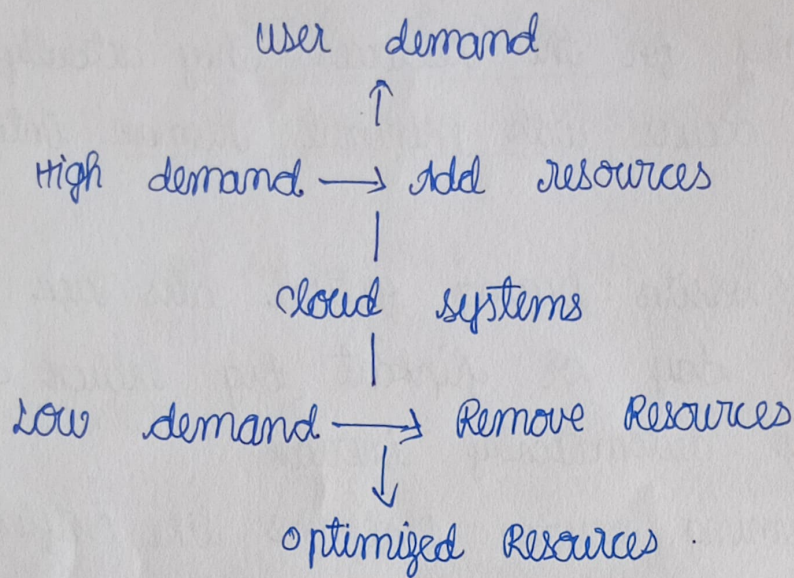
* Video streaming services: Platforms like Netflix scale resources during peak viewing hours.

* online learning platforms: Additional servers are allocated during online examinations or large live classes.

comparison of Traditional systems and Rapid Elasticity

Feature	Traditional infrastructure	cloud with Rapid Elasticity
Resource allocation	Fixed	Dynamic
Scalability	Limited	Automatic and Fast.
Cost	High due to over-provisioning.	Pay-as-you-use.
Performance during Peak Load	may degrade	Maintained efficiently.
Resource utilization	often underutilized	optimized.

Illustration .



Conclusion :-

Rapid elasticity enables cloud systems to automatically adjust resources according to workload requirements. It ensures scalability, cost efficiency, high performance, availability, making cloud computing highly suitable for applications.